

INTEGRATED SUMMATIVE ASSESSMENT: SUB-TASK 1 – WORKSHEET 1

COMPARISON OF COMPUTER PARTS

Processor Comparison

Feature	Processor A: AMD Athlon 3000G AM4	Processor B: Intel Celeron G5905 LGA1200
Manufacturer	AMD	Intel
Clock Speed	3.5GHz	3.5GHz
Cores	2	2
Price in Rand	R800.00	R750.00

Recommended Processor: AMD Athlon 3000G AM4

Motivation: Although R50 more expensive than the Intel Celeron G5905, the AMD Athlon 3000G includes integrated Vega graphics, removing the need to buy a separate graphics card. It uses Socket AM4, which is compatible with the selected ASUS A320M-K motherboard, keeping the build cost-effective for an entry-level Windows 10 Home system.

Motherboard Comparison

Feature	Motherboard A: ASUS A320M-K AM4	Motherboard B: Gigabyte A320M-S2H AM4
Manufacturer	ASUS	Gigabyte
RAM type	DDR4	DDR4
Price in Rand	R1,199.00	R1,099.00

Recommended Motherboard: ASUS A320M-K AM4

Motivation: The ASUS A320M-K uses Socket AM4, fully compatible with the selected AMD Athlon 3000G processor, and supports DDR4 RAM up to 2666MHz, matching the selected RAM. Choosing ASUS across the build also keeps the parts within one brand ecosystem, which can simplify driver support and warranty claims, even though it costs R100 more than the Gigabyte option.

RAM Comparison

Feature	RAM A: ASUS DDR4 4GB 2666MHz	RAM B: Kingston DDR4 4GB 2400MHz
Manufacturer	ASUS	Kingston
Type	DDR4	DDR4
Capacity	4GB	4GB
Price in Rand	R300.00	R280.00

Recommended RAM: ASUS DDR4 4GB 2666MHz

Motivation: The ASUS DDR4 2666MHz RAM matches the maximum supported speed of the ASUS A320M-K motherboard, giving slightly better performance than the 2400MHz Kingston option for only R20 more, while keeping the build consistent with the ASUS motherboard and processor pairing.

INTEGRATED SUMMATIVE ASSESSMENT: SUB-TASK 1 – WORKSHEET 2

COMPUTER COMPONENTS PRICES

COMPONENT	SELECTED PRODUCT	PRICE
Processor	AMD Athlon 3000G AM4	R800.00
Motherboard	ASUS A320M-K AM4	R1,199.00
RAM x 1	ASUS DDR4 4GB 2666MHz	R300.00
Hard drive x 1	ASUS 120GB SSD	R350.00
Power supply	ASUS 450W PSU	R500.00
Computer case	ASUS Basic Micro-ATX Case	R600.00
Monitor	ASUS VA195 19.5" HD Monitor	R1,500.00
Mouse	ASUS Basic Wired Mouse	R100.00
Keyboard	ASUS Basic Wired Keyboard	R150.00
Graphics card	Integrated (Athlon 3000G Vega)	R0.00
Speakers	ASUS USB 2.0 Speakers	R150.00
COST OF COMPUTER SYSTEM		R5,649.00

Note on quantities

Worksheet 2 lists single-unit prices (one processor, one motherboard, one RAM stick, one hard drive) to build one complete entry-level computer, giving a system cost of R5,649.00. If building two full computers instead (doubling the processor, motherboard, RAM and hard drive as originally researched), the total rises to R9,598.00, per the calculation below.

DESCRIPTION (build for 2 computers)	COST
Motherboard x2 (R1,199 x 2)	R2,398.00
Processor x2 (R800 x 2)	R1,600.00
RAM x4 (R300 x 4)	R1,200.00
Hard Drive x4 (R350 x 4)	R1,400.00
Power Supply x1	R500.00
Computer Case x1	R600.00
Total Core System Cost	R7,698.00
Total Peripherals Cost (Mouse, Keyboard, Monitor, Graphics Card, Speakers)	R1,900.00
GRAND TOTAL	R9,598.00

Compatibility Check & Motivation

1. Socket Compatibility: The AMD Athlon 3000G uses Socket AM4, which is fully compatible with the ASUS A320M-K motherboard.
2. RAM Compatibility: The ASUS A320M-K supports DDR4 RAM up to 2666MHz, which matches the selected ASUS DDR4 4GB 2666MHz RAM.
3. Power Requirement: The 450W ASUS PSU provides sufficient power for the 35W Athlon 3000G processor and all other components.
4. Budget Friendly: This is the cheapest working configuration using only ASUS parts that can run Windows 10 Home for basic office tasks.